Average Temperature of Cali, Colombia

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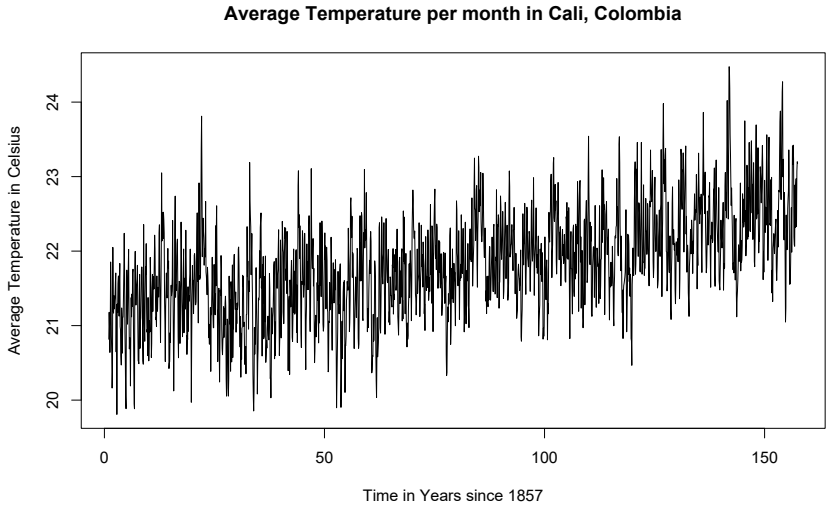

Figure 1: Cleaned data

Dataset

The data used in this project was from Kaggle, and represents the average global land temperatures (in degrees Celsius) for cities that are considered "major" from the years 1850 until 2013, recorded by month. The city we chose to look at was Cali, the most populous city in southwest Colombia.

The data was provided in a .csv file, with 239,177 rows and 7 columns in total. However, since we chose to focus on just Cali, we filtered this data to only look at the rows in which "Cali" was the city, and since the other columns aside from average temperature were irrelevant to time series (such as latitude and longitude), we dropped those as well. After doing this, we noticed that many of the earlier years consisted of NA values for the average temperature, so we created a function to find the max index of which NA was found, and remove everything up to that index. The end result was a data frame consisting of 1,879 rows with the date starting at February of 1857, which we then cast to a time series object with frequency of 12 to have the months represented properly.

A plot of the cleaned and finished data can be seen at the top of the page in Figure 1. The horizontal axis represents the years since 1857, and the vertical axis represents the average temperature in Celsius of Cali. At a quick glance, we can see this data is trending upwards fairly linearly. Why this matters will be talked about later on in this report, but this shows that over time (many years), the temperature of Cali is steadily increasing. Keep in mind that this is over quite a long period of time; more than one's lifetime, so it will be unlikely that we can see this upward trend in a matter of a few years, but over a period of about 50 years, we can notice the difference of average temperature.

Our goal in this report is to try our best to fit a model that can forecast the temperature of Cali over some time period (the further out we go, the harder this gets). This is an important topic in many areas due to global warming having a direct effect not just on the land itself, but the environment and inhabitants of the area (be it city, country, state, etc).

Model Selection

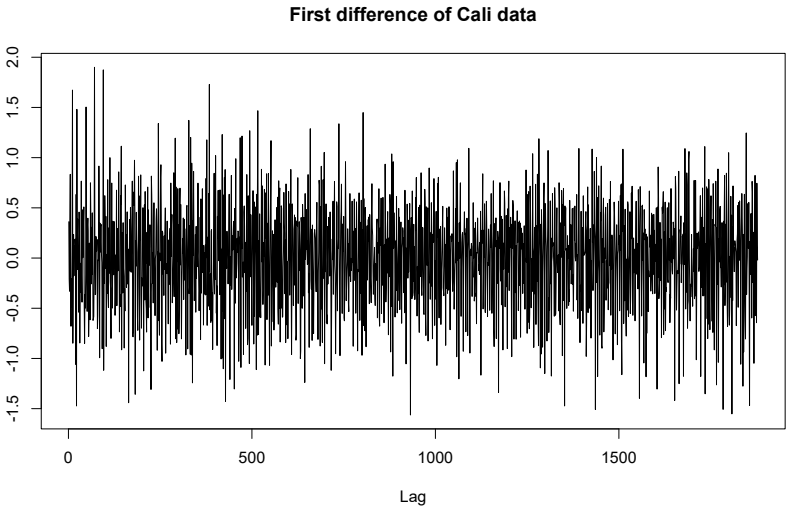

Figure 2: First Difference of Cali Data, x_{t-1}

As seen in Figure 1, there was a fairly linear trend to the Cali data. Because of this, we decided to try differencing the Cali data to see if it became stationary. As we can see in Figure 2, the time series now seems stationary. We then decided to take a look at the ACF of the differenced time series, which can be seen below in Figure 3.

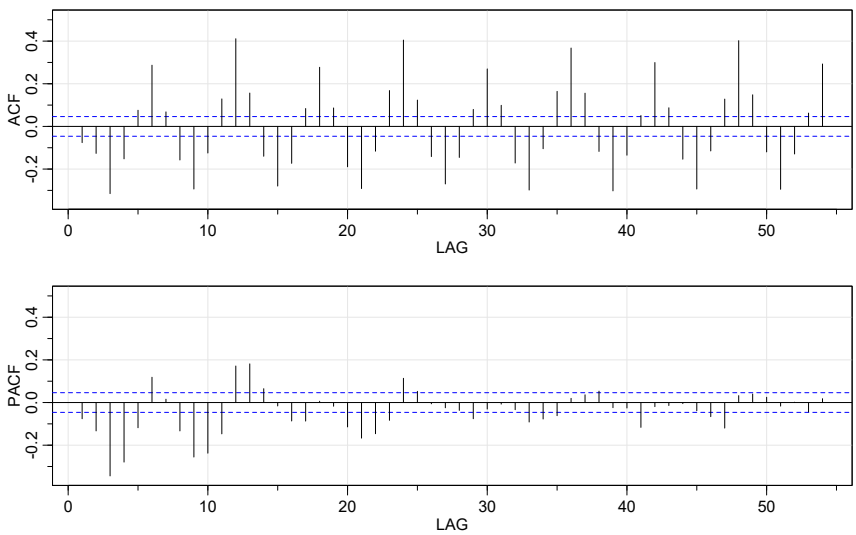

Figure 3: ACF and PACF of the first difference of Cali Data

This ACF indicates that there is some seasonal component that is missing here. So we discarded our first differenced data, and took the 12th difference instead, which resulted in what seems to be another stationary process. The plot of the seasonally differenced data and the ACF/PACF of said data can be found in Figure 4. The result still seems to be a stationary process, and as we can see from the ACF/PACF, the seasonality seems to be taken care of. The patterns of the ACF/PACF seem to be somewhat irregular from what we have learned, but came up with multiple models to try.

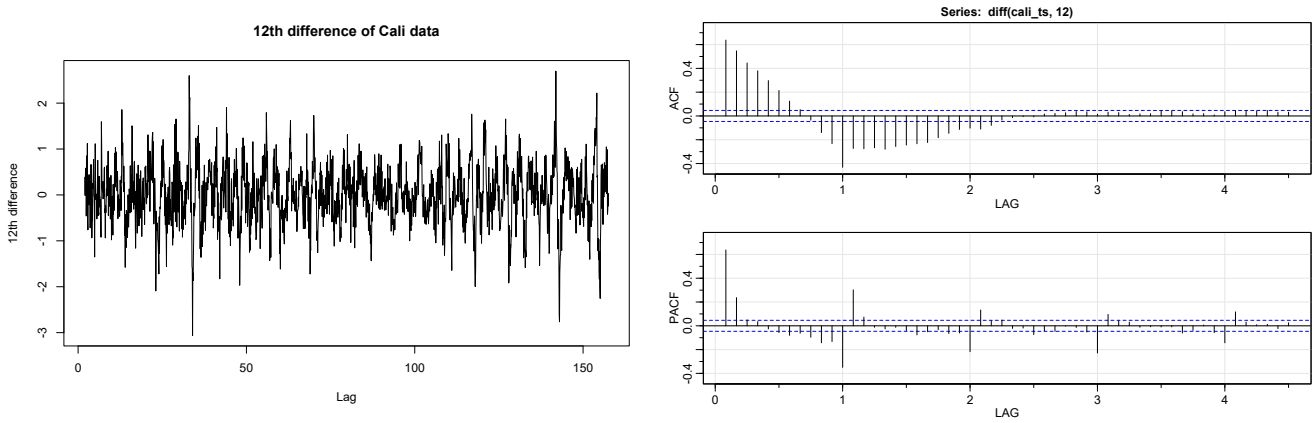

Figure 4: Seasonal Difference of Cali and the corresponding ACF/PACF.

Referencing the ACF/PACF of Figure 4, the models we came up with can be found in Table 1. Models 1 through 8 were determined from the ACF/PACF and possible subsets from those plots. Model 9 came from the fact that when we ran `sarima(...)` on the same parameters from model 4 (best choice on basis of AIC), the p-value of the AR(2) component was largely insignificant as can be seen in Figure 5. When cutting that term out, we came up with model 9, which gave a slightly better AIC, but a big improvement in the BIC.

```
         Estimate     SE   t.value p.value
ar1        0.8131 0.0833    9.7650  0.0000
ar2        0.0345 0.0610    0.5665  0.5711
ma1       -0.3371 0.0808   -4.1744  0.0000
sma1      -0.9592 0.0085 -112.2765  0.0000
constant   0.0007 0.0001    5.1267  0.0000
```

Figure 5: Coefficients and p-values for `(m4)`

Model	ARIMA with S=12	AIC	BIC	Passed Ljung Test?
`(m1)`	$(2,0,1)(0,1,2)$	1430.52	1463.71	X
`(m2)`	$(2,0,1)(1,1,1)$	1430.52	1463.71	X
`(m3)`	$(2,0,1)(1,1,2)$	1432.42	1471.15	X
`(m4)`	$(2,0,1)(0,1,1)$	1428.55	1456.21	X
`(m5)`	$(2,0,0)(0,1,2)$	1445.40	1473.06	X
`(m6)`	$(2,0,0)(1,1,1)$	1445.40	1473.06	X
`(m7)`	$(2,0,0)(1,1,2)$	1447.54	1480.73	X
`(m8)`	$(2,0,0)(0,1,1)$	1443.57	1465.70	X
`(m9)`	$(1,0,1)(0,1,1)$	1426.62	1448.75	X
`(m10)`	$(1,0,1)(2,1,0)$	1867.81	1895.48	X
`(m11)`	$\text{ARMA}(1,1)+\text{GARCH}(1,1)$	2533.596	-	X
`(m12)`	$\text{ARMA}(1,13)+\text{GARCH}(1,1)$	1389.885	-	✓

Table 1: Potential models

Then, after all the previous models, using the `auto.arima` command in `R` gave the resulting model as `(m10)`. What was surprising was the fact that the AIC of the model from `auto.arima` was actually the worst of all models we chose based on the plots of our ACF/PACF. We did not want to completely discard this model, so we continued the

model diagnostics using `(m10)` and `(m9)`, one from the `auto.arima` and one picked on the basis of best AIC.

Model Diagnostics

Model Diagnostics for both model 9 and model 10 from Table 1 were analyzed using the `sarima(...)` command in `R`. This command provided a plot of the standardized residuals (which should look like white noise), the ACF of the residuals, the normal Q-Q plot of the standardized residuals, and the p-values for the Ljung-Box statistic. The four figures can be found for each model in Figure 6 and Figure 7.

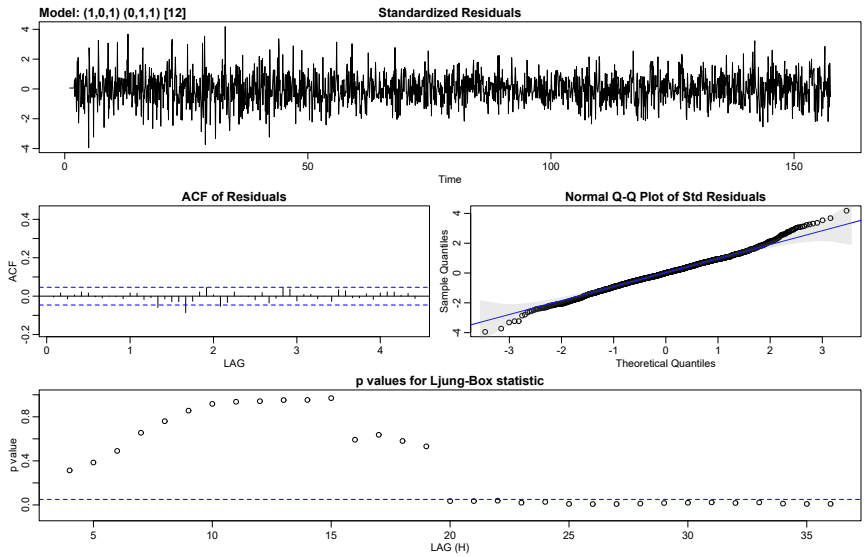

Figure 6: Model Diagnostics for Model 9

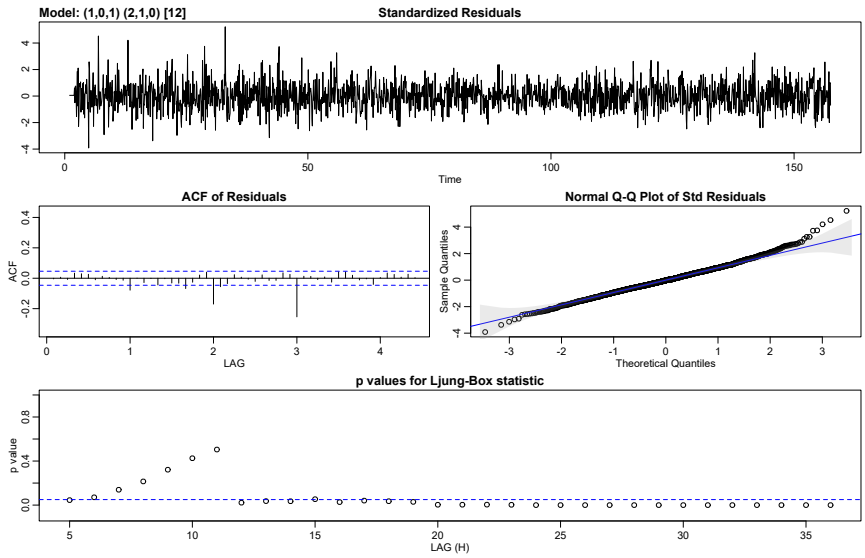

Figure 7: Model Diagnostics for Model 10

First looking at Figure 6, the diagnostics for model 9, the residuals look like white noise and the ACF of the residuals doesn't look too abnormal aside from some borderline

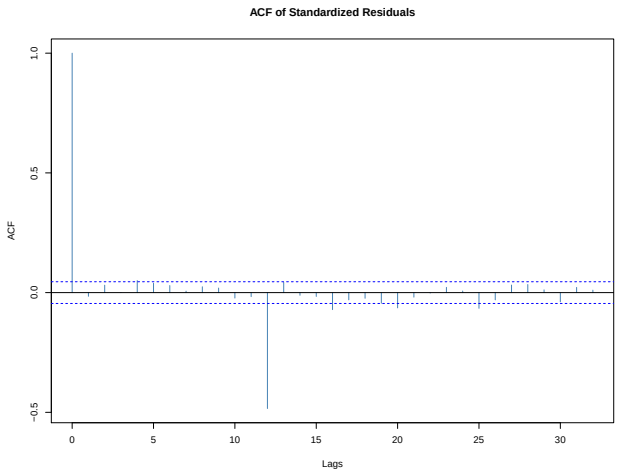

Figure 8: ACF of ARMA(1,1) + GARCH(1,1) on the seasonally differenced data.

cases towards the middle. The Q-Q Plot looks standard, but the p-values for the Ljung-Box statistic are a bit worrisome since after Lag $H = 20$ they do dip into significant territory, leaving us to reject the null hypothesis, where H_0 is the model being adequate. Nonetheless, this was the best model we came up with thus far on the basis of AIC, but we will also look at GARCH models coming up.

Next, we look at Figure 7 to see the diagnostics of model 10. Here, the residuals also look like white noise, but the ACF of the residuals is very concerning as the spike is significant at lags $H = 1, 2, 3$, where it gets stronger each time. The Q-Q Plot also seems to drift away towards the end, but seems fairly normal for the most part. Again, when looking at the p-values for the Ljung-Box statistic, we start off bad from the start where we are borderline significant. This does trend upwards going further and further from being a significant p-value though, until we hit Lag $H = 12$, which is the point where the p-values are significant from then on, suggesting an inadequate model. While unfortunate, we did decide to check out GARCH models, although our variance wasn't too spikey, just in case something may have popped up that we were missing. We also tried transforming the data via log-transform, but the results were actually similar at best, usually a bit worse, so we threw those considerations out.

GARCH Models

Although our experience with GARCH models is a bit limited and our data didn't seem to have enough spikes in the variance to justify these models (reference Figure 4), we decided to try them anyway since we couldn't come up with a great model to fit all of the assumptions. We essentially tried regular ARCH models, so ARCH(1,0) and ARCH(1,1), and then added the ARMA components to those to see what would come out best. In the end, we narrowed it down to the ARMA(1,1) + GARCH(1,1) model.

Unfortunately, the AIC for this was a terrible 2509.244, almost double that of our `(m9)`. We think this is because of the inability to combined a seasonal component with the GARCH model, so accounting for the seasonal component in our SARIMA models far outweighed what the GARCH model was doing to try and account for the spiking

```
Standardised Residuals Tests:
                                 Statistic p-Value
 Jarque-Bera Test   R    Chi^2  14.25     0.0008028
 Shapiro-Wilk Test  R    W      0.9969    0.0008388
 Ljung-Box Test     R    Q(10)  4.237     0.936
 Ljung-Box Test     R    Q(15)  7.652     0.937
 Ljung-Box Test     R    Q(20)  24.19     0.2344
 Ljung-Box Test     R^2  Q(10)  17.91     0.05651
 Ljung-Box Test     R^2  Q(15)  33.05     0.004624
 Ljung-Box Test     R^2  Q(20)  44.03     0.001491
 LM Arch Test       R    TR^2   23.63     0.02285
```

Figure 9: Standardized Residuals Tests for ARMA(1, 13) + GARCH(1, 1).

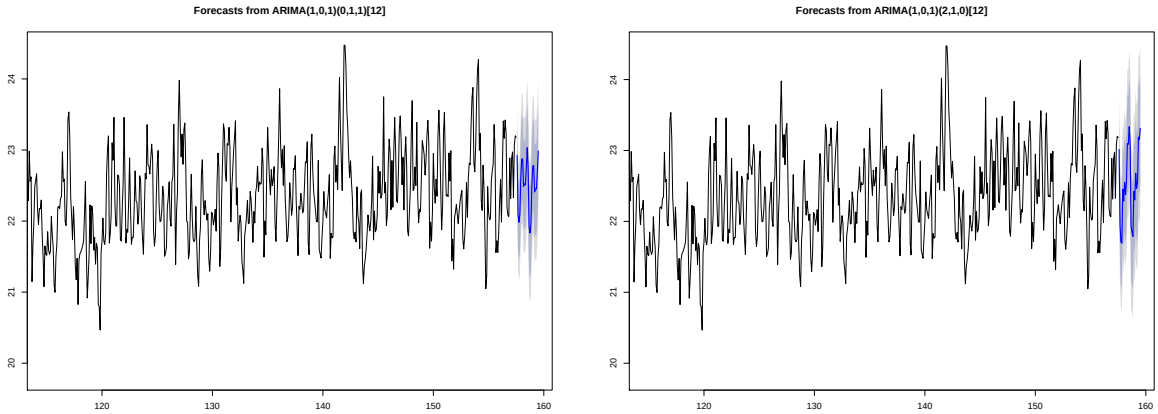

Figure 10: Forecasting for two years ahead using `(m9)` vs `(m10)`

variance.

What we had read up on doing was fitting an SARIMA model to the data, and fitting a GARCH model to the residuals. We tried to get this working in `R`, but were unable to find a way to do so. This did sound promising and something we would want to try and incorporate if we knew how.

Last, what we tried was to do an ARMA(1,13) + GARCH(1,1) model because of the spike in the ACF of our GARCH model which was seasonally differenced at lag $H = 12$, referenced in Figure 8. The MA(13) component was successful in getting rid of the spike, and actually turned out to have an even better AIC than our previous model, as can be seen in Table 1. Perhaps of more importance though is the fact that this model passed the Ljung-Box Statistic Test (Figure 9), saying that our model is adequate.

Forecasting

For forecasting, we will only look at `(m9)`, `(m10)` and `(m12)` from Table 1. Figure 10 shows the data from 1972 until 2013, and then forecasts for two years ahead up to 2015 (represented by the blue line with confidence bands around it). Model 10 seems to be going a bit upwards with a wider range in its forecasts, while model 9 seems to stay more towards the center of the data.

Since it is difficult to see clearly what is going on between the two models, we zoomed in to only show the years 2011 - 2013 with the forecasts still going two years ahead until 2015 in Figure 11. Now, it may be a bit more clear to see that model 9 stays more

towards the center and has a smoother curve overall, whereas model 10 displays very spiky patterns and has higher highs as well as lower lows.

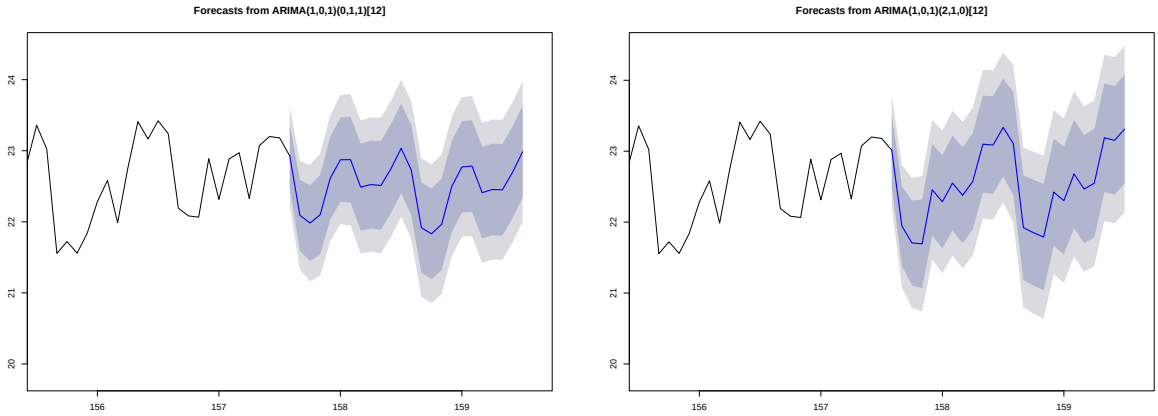

Figure 11: Zoomed in forecasting for two years ahead using `(m9)` vs `(m10)`

While it is hard to see a trend in two years for average temperature usually, we can sort of see the data predictions for model 10 getting slightly higher while the predictions for model 9 actually went a bit lower. This is interesting, because even though model 9 had a much lower AIC than model 10, model 10 seems to be capturing the trend that our eyes tell us is there from Figure 1 although it is slight, which model 9 does not.

`(m9)`		
ϕ_1	θ_1	Θ_1
0.8588	-0.3783	-0.9591

`(m10)`			
ϕ_1	θ_1	Φ_1	Φ_2
0.8647	-0.3959	-0.6486	-0.3073

`(m9)` without 2010-2013		
ϕ_1	θ_1	Θ_1
0.8802	-0.3977	-0.9428

`(m10)` without 2010-2013			
ϕ_1	θ_1	Φ_1	Φ_2
0.8664	-0.3966	-0.6472	-0.3078

Table 2: Coefficients for Models 9 and 10 with and without the 3 year cut.

For the next set of forecasts, we cut out the last three years of data from our data set and went through the same model selection process, which resulted in the same model parameters as `(m9)` and the same `(m10)` from `auto.arima`. We looked at the coefficients of the data after cutting out the last three years to see how similar they were to the models prior to the cutting, and they were very similar (see Table 2). We were able to see how our models using all the data would have performed if we only had the years up until 2010 as data, doing a sort of cross-validation of our models. These plots can be found in Figure 12 and Figure 13.

Now, we can see even more clearly over the period of three years that model 10 captures that slight positive linear trend that model 9 seems to be missing. When we predicted further ahead than four years, both model 9 and model 10 started to stay constant, leveling off at their mean perhaps.

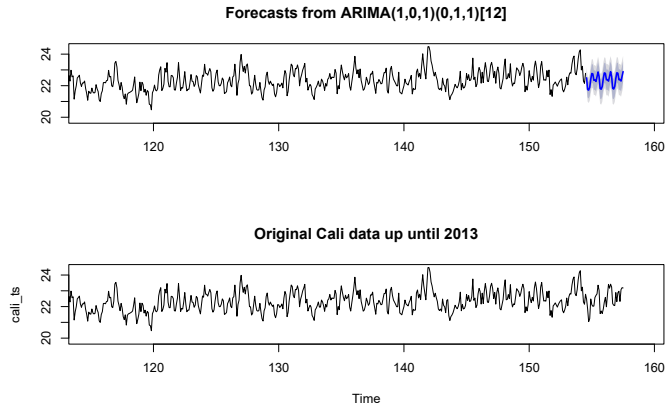

Figure 12: `(m9)` at the top from 1972 - 2010 with a 3-year forecast. Original data from 1972 - 2013 on the bottom.

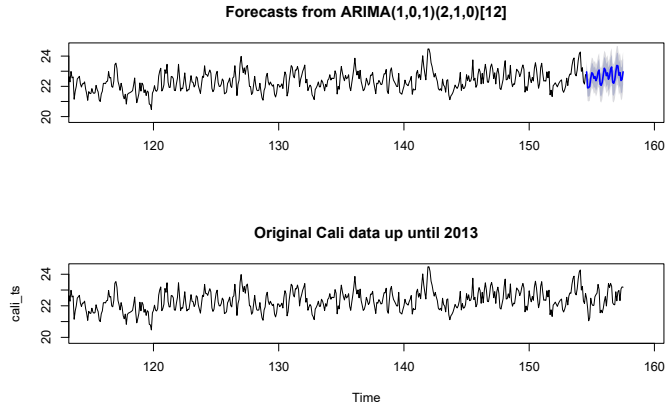

Figure 13: `(m10)` at the top from 1972 - 2010 with a 3-year forecast. Original data from 1972 - 2013 on the bottom.

The final forecast we did was on `(m12)` and can be seen in Figure 14. This one was mata bit trickier to do and interpret initially because we had to undo the difference of our time series and then predict. The green lines represent the 95% confidence intervals for the average temperature of Cali. What we notice immediately is that after 13 months, the trend goes upward with a slight curve, following what is assumed to be the trend and no longer using its past values.

Conclusion

Finishing up our model selection, we believe that our `(m12)` is the best models we could come up with given the tools at our disposal. Considering even the log transform gave worse results than just the seasonal difference, and the GARCH model passing the Ljung-Box Statistic Test while not accounting entirely for seasonality, we are happy with where we ended up.

On the sole basis of AIC, `(m12)` was the clear winner. When looking at the plots of our forecasts, `(m10)` and `(m12)` seemed to capture the slight positive linear trend better than `(m9)` did.

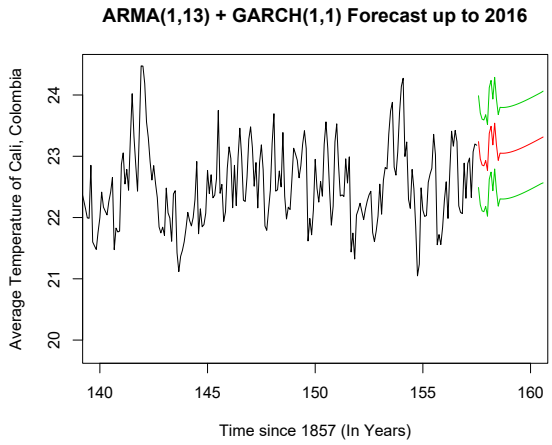

Figure 14: `(m12)`'s forecast up through 2016.

All things being considered, our final choice for a model would be that of `(m12)`, due to the fact that it seemed to forecast better than the other models did, had a better AIC, and passed the Ljung-Box Statistic Test where others failed. Ideally, at some point in the future, we would be able to fit a GARCH model that includes the seasonal component that weighs a lot in our data set, but for now we cannot. Our `(m12)` does its best to capture that, but it is still not done the way we wish it could be.